



Agricultural worker (crops)

Task A: Using conversion in the agricultural industry

1) Which farmer is getting the best price for oats?

\$1 = £0.78

Selling price per bushel = \$3.06

Selling price per tonne = £274

$$1 \div 0.014515 = 68.894...$$

$$\text{Price in dollars of a tonne} = 68.894... \times \$3.06 = \$210.816...$$

$$210.816... \times 0.78 \approx £164.44$$

The farmer in the UK is getting a better rate of pay.

2) \$1 = £0.78

Selling price per bushel = \$3.78

What is the minimum selling price per tonne of corn in the UK, to make the UK price better? Give your answer to the nearest pound.

$$1 \div 0.0254 = 39.37...$$

$$\text{Price in dollars of a tonne} = 39.37... \times \$3.78 = \$148.818...$$

$$148.818... \times 0.78 = £116.078...$$

£117 per tonne (needs to be more than £116)

3) \$1 = £0.78

Selling price per tonne = £153

What is the minimum selling price per bushel of soybeans in the US, to make the US price better? Give your answer to the nearest cent.

$$1 \div 0.027216 = 36.743...$$

$$\text{Price in dollars of a tonne} = 153 \div 0.78 = \$196.153...$$

$$\$196.153... \div 36.743... = \$5.3385...$$

\$5.34 per bushel

Crop	Bushel (bu) to tonne (t)
Barley	1 bu = 0.021772 t
Corn	1 bu = 0.0254 t
Oats	1 bu = 0.014515 t
Soybeans	1 bu = 0.027216 t
Wheat	1 bu = 0.027216 t



Task B: Using estimation in the agricultural industry

1) Calculate the total income from a 15 hectare field of oats.

Average mass of a grain = 41.3 mg

Cost per tonne = £274

Number of panicles per square metre samples:

412, 423, 404, 398 and 438

Number of grains per panicle:

43, 49, 41, 52 and 50

$$\text{Mean number of panicles} = 2075 \div 5 = 415$$

$$\text{Mean number of grains per panicle} = 235 \div 5 = 47$$

$$\text{Total number of grains} = 415 \times 47 \times 150\,000 = 2.92575 \times 10^9$$

$$\text{Total mass of grain} = 2.92575 \times 10^9 \times 41.3 \times 1 \times 10^{-9} = 120.8... \text{ tonnes}$$

$$\text{Total income} = 120.8... \times £274 \approx £33\,108.37$$

